

Assignment 1 : Understanding Users

1. Short Description of the Topic

Our project focusses on understanding how older adults interact with technology,

particularly mobile applications and identifying the challenges they face. The goal is to

explore usability barriers, accessibility preferences, and design improvements that

can make mobile application more inclusive and easier to use for demographic.

Older adults often face physical and cognitive challenges that affect their

interaction with technology. Understanding

their needs helps designers create more

intuitive and accessible products. The intended

user are individuals age 55 and above

who use smartphones, tablets, or computers

in their daily lives for communication, health management and entertainment.

2. Design of Semi-structured Qualitative Interview.

Main research areas

RQ1 - challenges faced by older adults that can be alleviated through technology.

RQ2 - How older adults interact with and prefer current technologies.

RQ3 - Features that improve the usability of technological products for older adults.

Interview structure

Duration 20-30 minutes

Format - Face to face or video call

subtopic - Daily activity, health management, social interaction, technology usage, design

preferences, accessibility features

Sample Questions

- What daily activities do you find challenging as you age?
- What difficulties do you encounter when managing your health?
- How do you stay connected with friends and family?
- What types of devices do you use the most and how did you learn to use them?
- What features or designs make a device easier or harder to use?

3. Recruitment of participants.

Two participants were selected using Purposive sampling to ensure they represented the target user group (older adults aged 65+)

Participants were approached personally through

Family connections and community centers.

Each was briefed about the research purpose and gave verbal consent before participation.

Participant - Female age 55 - 64, house wife

Participant - Male age 65 - 74, business.

4. Interview notes Summary

Participant 1 - uses smartphone primarily for youtube and messenger for video calls to family. Faces challenges with network issues.

Advertisements interrupting content and

small text requiring glasses. Learned by

observing others. Prefers voice based

interaction over typing

Participant 2 - uses button phone mainly for calls and balance checks. Occasionally smartphone for basic tasks. Avoid complex app. Messenger

or new features due to memory issues
and preference for simplicity sticks to
memorized sequences to avoid trouble.

5. Interview analysis

key takeaways

- Simplicity and readability and crucial
design factor with both participants
emphasizing larger button / text and
avoidance of complexity.

Participant 1 - Highlights interruption
and prefers voice for communication.

Participant 2 - Stresses minimalism, memorization
of basics and aversion to learning new
features due to cognitive load.

common themes : voice assistance valued over
typing frustration from unreliable network

balances for learning.

6. Contextual Enquiry

The contextual enquiry focused on how older adults use smartphones in their daily lives.

Participants were observed performing routine activities such as checking messages, viewing news and using video apps.

- Identify difficulties during real world app use.
- Observe physical interaction patterns.
- Record request or confusion point use.

~~Box~~ Duration - 25 minutes.

7. Contextual Enquiry Results

Participant 1 - Observed hesitation with small buttons and ads during youtube navigation. Frequent pauses to adjust glasses for text. Relied on voice calls in messenger but struggled with network.

Participant 2 - Quick with basic calls but avoided exploring new menus, misremembered sequences for balance checks, leading to frustration. Preferred verbal confirmation before actions.

8. Contextual Enquiry Analysis

Physical limitation (vision and hand stability) affect usability, as seen in needs for glasses and larger elements.

- Repeated logging requirement and hidden function cause confusion, aligning with avoidance of complexity.
- Visual clutter (ads) and poor contrast reduce com font.
- Older adults rely heavily on memory, repetition, and family help to learn interfaces with interruption exacerbating issues.

Contextual Function Analysis